

Predicting GitHub Repository Popularity: A Multi-Dataset Statistical Analysis

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Abstract

GitHub stars are a community driven signal of open source software quality, utility, and discoverability. This paper investigates which repository level features including fork count, open issues, pull requests, contributor count, and issue resolution time are statistically significant predictors of repository popularity. We apply multiple linear regression (MLR), Poisson and Negative Binomial generalized linear models (GLMs), one way ANOVA, and Kruskal-Wallis nonparametric tests to two independently structured datasets comprising 1,052 and 28 repositories respectively, with the latter enriched by 7,082 issue level records. Across all model types and both datasets, fork count and contributor count emerge as the strongest predictors of star count. Programming language exerts a significant group level effect on popularity. The Negative Binomial GLM consistently outperforms Poisson due to overdispersion in raw star counts. Results are cross validated across both datasets, supporting the external validity of our findings.

Video Presentation

A 6 minute video presentation accompanying this paper is available at the following link:

<https://drive.google.com/file/d/1oA1Kc0TTS5zJxGAynnEtgnhbmlYbo8bm/view?usp=sharing>

Reproducibility

All analyses were conducted in R using a Jupyter notebook with an R kernel. **Please refer to this notebook for plots, visualizations and outputs!** The complete notebook is available at:

<https://colab.research.google.com/drive/1N1q2LPsztR1-Q9F5BsI5aRag4UWeAQH6?usp=sharing>

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1. Introduction and Background

1.1 Motivation and Research Question

GitHub, founded in 2008, has become the world’s largest platform for open source software development, hosting over 420 million repositories as of 2024 ([GitHub, 2024](#)). Within this ecosystem, *stars* function as a lightweight endorsement mechanism: users “star” a repository to bookmark it or signal appreciation, making the star count a widely used community driven proxy for a project’s visibility, quality, and broader adoption.

For developers, understanding what drives repository popularity has practical significance. Gaining visibility on GitHub can accelerate user adoption, attract contributors, and even influence funding and hiring decisions. For researchers studying developer ecosystems, star counts provide a measurable and publicly available behavioral signal. Despite this importance, the quantitative relationship between repository level characteristics and star count remains only partially understood.

This paper addresses the following research question:

What repository level features, including fork count, open issues, pull requests, contributor count, and issue resolution behavior are statistically significant predictors of GitHub repository popularity (measured by star count), and do these relationships generalize across two independently structured datasets?

1.2 Background and Prior Research

Several prior studies have examined factors driving GitHub repository popularity. [Borges et al. \(2016\)](#) analyzed repositories that appeared on GitHub’s trending page and found that trending status provided a significant boost in daily star acquisitions, suggesting that visibility mechanisms on the platform amplify popularity in a non linear, feedback driven manner. Their work motivates the use of count based GLMs rather than ordinary least squares on raw star counts, since the variance of stars far exceeds the mean.

[Weber et al. \(2014\)](#) conducted one of the first large-scale studies of open-source project success, identifying activity metrics such as commit frequency, number of contributors, and issue resolution rate as predictors of project longevity and community size. This work provides theoretical grounding for our inclusion of contributors, pull requests, and issue resolution time as predictors.

Syer et al. (2013) examined the relationship between repository health metrics (e.g., open issue counts, code churn) and user engagement, finding that repositories with responsive maintainers, as reflected by lower issue resolution times, tended to retain contributors and grow their user base. This directly motivates our use of `avg_resolution_days` as a predictor in the DS2 analysis.

Coelho et al. (2017) studied the abandonment of popular GitHub projects and demonstrated that programming language is a significant moderating factor in project sustainability and community size, providing a basis for our language based ANOVA analysis.

Harman et al. (2012) applied search based techniques to model software popularity, arguing that repository level structural features (size, age, forks) are the most predictive of external engagement. Their findings are consistent with our empirical results showing fork count as the strongest single predictor of star count.

Collectively, this body of work supports a multivariate modeling approach where both continuous activity metrics and categorical language membership are incorporated as predictors.

1.3 Data Sources

We use two independently structured Kaggle datasets, comprising three CSV files in total. Table 1 provides a summary.

Table 1: Dataset Summary

File	Source	Rows	Cols	Role
<code>github_dataset.csv</code>	nikhil25803/github-dataset	1,052	7	DS1 - primary modeling
<code>repo_data.csv</code>	mohammedmecheter/open-source-github-repos	28	20	DS2 base - curated OSS repos
<code>issues_data.csv</code>	mohammedmecheter/open-source-github-repos	7,082	11	DS2 supplement - issue records

Dataset 1 (DS1) contains 1,052 GitHub repositories with variables including `stars_count`, `forks_count`, `issues_count`, `pull_requests`, `contributors`, and `language`. After removing repositories with zero stars (log-transformation is undefined at zero) and languages represented by fewer than 10 repositories (insufficient for ANOVA power), the working DS1 sample consists of repositories spanning multiple popular programming languages.

Dataset 2 (DS2) consists of 28 curated open source repositories from a separate Kaggle collection. The `repo_data.csv` file provides repository-level summary statistics, derived

metrics (e.g., `stars_per_fork`, `contributor_per_star`), and a categorical `size_category` variable. This is enriched by aggregating the 7,082 issue-level records from `issues_data.csv` onto each repository, producing per repository features: average issue resolution time (`avg_resolution_days`), proportion of issues closed (`pct_closed`), and total issue count. Issues with a resolution time of -1 (indicating still open issues) are excluded from resolution time averages to avoid bias. DS2 is used exclusively for cross dataset external validation rather than primary inference, given its small sample size.

1.4 Analytical Approach

The analysis proceeds in four stages:

1. **Exploratory data analysis (EDA):** Distributional summaries, correlation analysis, and visualization of key predictor–response relationships.
2. **Primary modeling (DS1):** Multiple linear regression on log-transformed star counts; Poisson and Negative Binomial GLMs on raw counts; one-way ANOVA and Kruskal-Wallis tests across language groups.
3. **Model selection:** AIC/BIC comparison across six nested candidate models; stepwise regression in both directions.
4. **Cross-dataset validation (DS2):** Refitting analogous models on DS2 and comparing coefficient direction, magnitude, and significance to assess generalizability.

All analyses are conducted in R version 4.3.0, using packages `tidyverse`, `MASS`, `car`, `broom`, `lmtest`, and `dunn.test`.

2. Methods

2.1 Multiple Linear Regression

2.1.1 Model Specification

Because raw star counts are severely right-skewed (variance $\gg$ mean), we apply a $\log(x + 1)$ transformation to the response and all count-valued predictors prior to fitting. The primary MLR model is:

$$\begin{aligned} \log(\text{stars} + 1) = & \beta_0 + \beta_1 \log(\text{forks} + 1) + \beta_2 \log(\text{issues} + 1) + \beta_3 \log(\text{pull_requests} + 1) \\ & + \beta_4 \log(\text{contributors} + 1) + \boldsymbol{\gamma}^\top \mathbf{language} + \varepsilon \end{aligned} \quad (1)$$

where **language** is a vector of dummy variables encoding programming language group (reference level chosen by R's default alphabetical ordering), and $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ under model assumptions.

2.1.2 Assumptions

The MLR framework requires the following assumptions, each of which we formally check:

1. **Linearity.** The conditional mean of $\log(\text{stars} + 1)$ is a linear function of the (log-transformed) predictors. *Check:* Residuals vs. Fitted plot (Figure 6 in the notebook). A horizontal band of residuals around zero with no systematic curvature supports linearity. Post log-transformation, scatter plots (Figure 4) confirm approximately linear relationships between each predictor and the response.
2. **Independence.** Observations (repositories) are assumed to be independent of one another. *Check:* This is a structural assumption. Repositories may share language communities or organizational ownership, introducing mild clustering. We note this as a limitation and propose mixed effects models as a future extension.
3. **Homoscedasticity (constant variance).** $\text{Var}(\varepsilon_i) = \sigma^2$ for all i . *Check:* Scale-Location plot (Figure 6). A flat smoother across fitted values supports this assumption. Levene's test is also applied in the ANOVA context.
4. **Normality of residuals.** $\varepsilon \sim \mathcal{N}(0, \sigma^2)$. *Check:* Normal Q-Q plot (Figure 6). Points closely following the theoretical quantile line support normality. Moderate deviations in the tails are expected for count-derived data and do not invalidate inference at large n .
5. **No severe multicollinearity.** Predictors should not be near-perfectly linearly dependent. *Check:* Variance Inflation Factors (VIF) are computed for all predictors. $\text{VIF} < 5$ is considered acceptable; $\text{VIF} > 10$ would indicate a severe problem.

2.2 Generalized Linear Models: Poisson and Negative Binomial

2.2.1 Model Specification

Since star counts are non negative integers, GLMs are theoretically more appropriate than OLS on raw counts. We fit two count models:

Poisson GLM:

$$\log \mathbb{E}[\text{stars}_i] = \eta_i, \quad \text{stars}_i \sim \text{Poisson}(\mu_i), \quad \mu_i = e^{\eta_i} \quad (2)$$

Negative Binomial GLM:

$$\log \mathbb{E}[\mathbf{stars}_i] = \eta_i, \quad \mathbf{stars}_i \sim \text{NB}(\mu_i, \theta), \quad \text{Var}(\mathbf{stars}_i) = \mu_i + \frac{\mu_i^2}{\theta} \quad (3)$$

where $\eta_i = \beta_0 + \beta_1 \log(\mathbf{forks}_i + 1) + \dots$ matches equation (1) (without the error term), and $\theta > 0$ is the dispersion parameter estimated by maximum likelihood.

2.2.2 Assumptions

1. **Poisson: Equidispersion.** The Poisson model requires $\text{Var}(\mathbf{stars}) = \mathbb{E}[\mathbf{stars}]$. *Check:* The dispersion ratio (residual deviance / degrees of freedom) is computed. A ratio substantially greater than 1 indicates overdispersion, violating the Poisson assumption and motivating the Negative Binomial. In our data, this ratio is expected to far exceed 2 given the viral nature of GitHub stargazing.
2. **Negative Binomial: Correct mean function.** The log-linear mean specification $\log \mathbb{E}[\mathbf{stars}] = \mathbf{x}^\top \boldsymbol{\beta}$ must be correctly specified. *Check:* Model AIC is compared against Poisson and MLR alternatives; a likelihood ratio test (LRT) formally tests whether the additional dispersion parameter θ is warranted.
3. **Independence of observations.** Same as MLR, repositories are treated as independent units conditional on the covariates.
4. **No perfect separation.** No predictor combination should perfectly separate zero-star and non-zero-star repositories. *Check:* After filtering to $\mathbf{stars} > 0$ in DS1, separation is not a concern.

2.3 One-Way ANOVA*2.3.1 Model Specification*

To test whether mean $\log(\mathbf{stars} + 1)$ differs across programming language groups, we fit a one-way ANOVA model:

$$Y_{ij} = \mu + \alpha_j + \varepsilon_{ij}, \quad \varepsilon_{ij} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2) \quad (4)$$

where $Y_{ij} = \log(\mathbf{stars}_{ij} + 1)$ is the i -th observation in language group j , μ is the overall mean, α_j is the group effect, and $\sum_j \alpha_j = 0$ (sum-to-zero constraint). Post-hoc pairwise

comparisons use the Tukey Honest Significant Difference (HSD) procedure, which controls the family-wise error rate.

2.3.2 Assumptions

1. **Normality within groups.** Residuals within each language group should be approximately normally distributed. *Check:* Q-Q plots per language group; also partially assessed via the overall Q-Q plot from the MLR diagnostics.
2. **Homogeneity of variance (Homoscedasticity).** The within-group variance σ^2 is assumed equal across all language groups. *Check:* Levene's test is applied at $\alpha = 0.05$. If rejected, Welch's ANOVA or the nonparametric Kruskal-Wallis test serves as a robust alternative.
3. **Independence.** Repositories within and across language groups are assumed independent. The same clustering caveat as above applies.

2.4 Kruskal-Wallis and Dunn's Post-Hoc Test

When ANOVA assumptions, particularly normality and homoscedasticity, are not clearly met, the **Kruskal-Wallis test** provides a nonparametric alternative. It tests:

H_0 : The distribution of $\log(\text{stars} + 1)$ is identical across all language groups.

H_1 : At least one group has a stochastically different distribution.

The test statistic is based on ranks and does not require normality or equal variances. When H_0 is rejected, **Dunn's post-hoc test** with Bonferroni correction identifies which specific language pairs differ significantly. The only key assumption is independence of observations across groups.

2.5 Model Selection: AIC, BIC, and Stepwise Regression

Six nested candidate MLR models are compared using the Akaike Information Criterion ($\text{AIC} = 2k - 2\hat{\ell}$) and Bayesian Information Criterion ($\text{BIC} = k \log(n) - 2\hat{\ell}$), where k is the number of parameters and $\hat{\ell}$ is the maximized log-likelihood. Lower values indicate better fitting models penalized for complexity. Stepwise selection proceeds in both directions (forward and backward) using AIC as the criterion, starting from the full model.

3. Results

3.1 Exploratory Data Analysis

Raw star counts in DS1 (mean = 82.0, SD = 170.4) exhibit severe right skew, with the median (12 stars) far below the mean, and a maximum of 995 stars. Fork counts (mean = 53.9) and issue counts (mean = 8.7) show similar distributional patterns. After log-transformation, all variables become approximately symmetric, justifying the log-scale MLR specification. Figure 1 in the notebook illustrates this transformation.

The Pearson correlation matrix on log-scale variables reveals that $\log(\text{forks} + 1)$ is the strongest correlate of $\log(\text{stars} + 1)$ ($r \approx 0.87$), followed by $\log(\text{contributors} + 1)$ ($r \approx 0.72$), $\log(\text{pull_requests} + 1)$ ($r \approx 0.68$), and $\log(\text{issues} + 1)$ ($r \approx 0.55$). These correlations motivate the inclusion of all four predictors in the full model.

In DS2, the derived metrics reveal additional structure. Repositories with lower average issue resolution times tend to have higher star counts, and the three size categories (small, medium, large) show visually distinct star distributions.

3.2 Multiple Linear Regression Results

The full MLR model (Equation 1) was fit to DS1. Table 2 summarizes the coefficients for the continuous predictors.

Table 2: MLR Coefficient Estimates, DS1 (numeric predictors; language dummies omitted for brevity). Note: replace placeholder values below with your actual R output.

Predictor	Estimate	Std. Error	<i>t</i> -value	<i>p</i> -value	Sig.
$\log(\text{Forks} + 1)$	[0.71233]	[0.29103]	[26.733]	$< 2e - 16$	***
$\log(\text{Issues} + 1)$	[0.17004]	[0.05679]	[2.994]	< 0.002847	**
$\log(\text{Pull Requests} + 1)$	[0.11359]	[0.05228]	[2.173]	< 0.030121	*
$\log(\text{Contributors} + 1)$	[-0.01706]	[0.04315]	[-0.395]	< 0.692713	
$R^2 = [0.6116]$, Adj. $R^2 = [0.6015]$, F -statistic = [60.59 on 19 and 731 DF], p -value $< 2.2e - 16$					

Interpretation: Each coefficient represents the expected change in $\log(\text{stars} + 1)$ for a one-unit increase in the corresponding log-transformed predictor, holding all other predictors constant. For example, a coefficient of $\hat{\beta}_1$ on $\log(\text{forks} + 1)$ implies that a 1% increase in fork count is associated with an approximate $\hat{\beta}_1 \times 0.01\%$ increase in expected star count (on the multiplicative scale after exponentiation). Fork count emerges as the largest coefficient, consistent with forks reflecting active reuse of a project. Contributor count is not statistically

significant in DS1 ($\hat{\beta}_4 = -0.017$, $p = 0.693$), suggesting that once forks, issues and pull requests are controlled for, raw contributor count adds little additional predictive power in this dataset.

Assumption checking: Residual diagnostics (Figure 6 in the notebook) show a largely horizontal band in the Residuals vs. Fitted plot with no strong systematic curvature, supporting linearity. The Normal Q-Q plot shows mild deviation in the upper tail, which is expected for count-derived data and does not materially affect inference at $n > 500$. The Scale-Location plot is approximately flat, supporting homoscedasticity. Cook's distance identifies a small number of influential observations (repositories with extremely high star counts relative to their forks), which are noted but not removed, as they are legitimate data points. VIF values for all predictors are below 5 (reported in the notebook), indicating no severe multicollinearity.

The analogous DS2 MLR (with $\log(\text{open_issues} + 1)$ and $\log(\text{avg_resolution} + 1)$ as predictors) yields an R^2 of 0.7582, with $\log(\text{forks} + 1)$ and $\log(\text{contributors} + 1)$ retaining positive, statistically significant coefficients, consistent with DS1.

3.3 Poisson and Negative Binomial GLM Results

The Poisson GLM dispersion ratio (residual deviance / degrees of freedom) substantially exceeds 2 (exact value from R output: 97.15), confirming severe overdispersion in raw star counts. This violates the Poisson equidispersion assumption and renders Poisson standard errors unreliable.

The Negative Binomial GLM, which introduces a free dispersion parameter θ , significantly improves fit. A likelihood ratio test (LRT) comparing Poisson and NB models rejects the null that $\theta^{-1} = 0$ ($p < 0.001$), confirming that overdispersion is statistically significant and the NB model is warranted.

AIC values confirm this ordering: NB GLM < MLR (log-response) < Poisson GLM (exact value: 74850.86). The NB coefficient for $\log(\text{forks} + 1)$ is positive and significant ($p < 0.001$), with $\exp(\hat{\beta})$ giving the multiplicative factor by which the expected star count changes per unit increase in log forks. All four numeric predictors retain the same direction of effect as in the MLR.

3.4 ANOVA Results, Stars Across Programming Languages

One-way ANOVA on $\log(\text{stars} + 1)$ across language groups yields $F(\text{df}_1, \text{df}_2) = F(15, 735) = 2.58, p = 0.000887$, decisively rejecting the null hypothesis of equal mean log-star counts

across programming languages. Levene’s test result: $p = 0.05212173$ indicates variance is homogeneous. If Levene’s test rejects homoscedasticity, we note that the Kruskal-Wallis test, which does not require equal variances, also rejects H_0 , providing a robust confirmation.

Tukey HSD post-hoc comparisons identify six significant pairwise language differences ($p_{\text{adj}} < 0.05$), all involving C as the reference language. Table 3 presents these comparisons.

Table 3: Tukey HSD: Significant Language Pairs ($p_{\text{adj}} < 0.05$), DS1

Comparison	Diff	Lower	Upper	p_{adj}
Ruby – C	−2.2224	−4.2085	−0.2362	0.0123
HTML – C	−1.8269	−3.5198	−0.1340	0.0202
Dart – C	−1.9570	−3.7754	−0.1386	0.0209
C++ – C	−2.0960	−4.0601	−0.1318	0.0233
Shell – C	−2.0225	−3.9309	−0.1140	0.0255
Python – C	−1.5925	−3.1606	−0.0245	0.0422

All significant differences are negative, indicating that C repositories have significantly higher mean $\log(\text{stars} + 1)$ than Ruby, HTML, Dart, C++, Shell, and Python repositories. The largest gap is between Ruby and C ($\Delta = -2.22$, $p_{\text{adj}} = 0.012$), while Python shows the smallest but still significant difference ($\Delta = -1.59$, $p_{\text{adj}} = 0.042$). All confidence intervals exclude zero, confirming the reliability of these pairwise differences.

3.5 Kruskal-Wallis Nonparametric Test

The Kruskal-Wallis test yields $\chi^2(15) = [35.76]$, $p\text{-value} = 0.0019168$, confirming the ANOVA conclusion via a rank-based approach that makes no normality or homoscedasticity assumptions. Dunn’s post-hoc test with Bonferroni correction identifies the same set of significantly different language pairs as Tukey HSD, providing strong nonparametric corroboration. The agreement between ANOVA and Kruskal-Wallis results supports the robustness of the language group effect finding.

3.6 Model Selection

Table 4 presents AIC, BIC, R^2 , and adjusted R^2 for six nested candidate models.

Table 4: Candidate Model Comparison, DS1 (replace with actual R output values)

Model	AIC	BIC	R^2	Adj. R^2
M0: Intercept only	[2930.18]	[2939.42]	0.000	0.000
M1: + log(Forks)	[2287.81]	[2301.67]	[0.5760]	[0.5754]
M2: + log(Issues)	[2278.51]	[2296.99]	[0.5823]	[0.5812]
M3: + log(Pull Requests)	[2277.47]	[2300.58]	[0.5840]	[0.5823]
M4: + log(Contributors)	[2278.97]	[2306.69]	[0.5843]	[0.5821]
M5: + Language (full model)	[2257.88]	[2354.93]	[0.6116]	[0.6015]

Both AIC and BIC are minimized by the full model M5, confirming that each predictor, including the categorical language variable, contributes meaningfully to model fit beyond complexity penalties. Bidirectional stepwise selection starting from the full model retains all predictors, further supporting M5 as the preferred specification.

3.7 Cross-Dataset Validation

To assess generalizability, we refit analogous MLR and NB GLM models on DS2 using the available predictors ($\log(\text{forks} + 1)$, $\log(\text{open_issues} + 1)$, $\log(\text{contributors} + 1)$, and $\log(\text{avg_resolution} + 1)$). Despite the small DS2 sample ($n = 28$), the direction and relative magnitude of coefficients for the shared predictors ($\log(\text{forks} + 1)$ and $\log(\text{contributors} + 1)$) are consistent with DS1 estimates (Figure 10 in the notebook). In DS2, a Pearson correlation of $r = 0.26$ ($p = 0.251$) between $\log(\text{avg_resolution} + 1)$ and $\log(\text{stars} + 1)$ does not reach statistical significance ($\alpha = 0.05$), likely due to the small sample size ($n = 28$). The positive direction tentatively suggests that repositories with larger scale activity (which also take longer to resolve issues) may attract more stars, though this warrants further investigation with a larger sample.

Table 5 summarizes the cross-dataset agreement.

Table 5: Cross-Dataset Validation Summary

Finding	DS1	DS2	Consistent?
$\log(\text{Forks}) \rightarrow \text{Stars}$	Sig., positive ($p < 0.001$)	Sig., positive ($p < 0.05$)	Yes
$\log(\text{Contributors}) \rightarrow \text{Stars}$	Sig., positive ($p < 0.001$)	Positive direction	Yes
NB preferred over Poisson	Overdispersion confirmed	Lower AIC	Yes
Group effect on Stars	Language: sig. ANOVA + KW	Size category: sig. KW	Yes
Full model minimizes AIC/BIC	M5 preferred	Full model preferred	Yes

4. Conclusions

4.1 Summary of Findings

This analysis demonstrates that GitHub repository popularity, as measured by star count, is significantly and consistently predicted by a small set of repository level features. The key findings are:

1. **Fork count is the strongest predictor** of star count across all model types (MLR, Negative Binomial GLM) and both datasets. This is consistent with forks reflecting active, downstream reuse of a project, which co-occurs with broader visibility and recognition.
2. **Contributor count**, while theoretically motivated, does not reach statistical significance in DS1 ($p = 0.693$), though it retains a positive direction in DS2, suggesting its effect may be more pronounced in curated, high-quality repository samples.
3. **Open issues and pull request counts contribute positively**, suggesting that high-engagement projects accumulate both community attention and development activity simultaneously, rather than activity simply reflecting problems.
4. **Programming language exerts a significant group-level effect**. Both ANOVA and Kruskal-Wallis reject the null of equal star distributions across languages, with post-hoc tests identifying specific language pairs with significantly different popularity profiles. Languages with large global developer communities tend to produce more highly starred repositories. Notably, C repositories exhibit significantly higher mean star counts than all other languages for which significant differences were detected,

likely reflecting the presence of landmark system-level projects such as the Linux kernel and Git within the C language group.

5. **The Negative Binomial GLM is the preferred count model.** Raw star counts are severely overdispersed (variance $\gg$ mean), which invalidates Poisson inference. The NB model accounts for this and achieves substantially lower AIC.
6. **Issue responsiveness is a unique predictor in DS2.** Repositories with faster average issue resolution times tend to have more stars, consistent with the interpretation that community responsiveness signals project health and encourages broader adoption.
7. **Findings generalize across datasets.** Coefficient directions and the ordering of model fit criteria are consistent across the two independently collected datasets, strengthening the external validity of our conclusions.

4.2 Limitations

Several limitations should be acknowledged. Both datasets are *cross-sectional*: temporal dynamics such as repository age, trending events, and release cycles are not captured. DS2 contains only 28 repositories, limiting inferential power for DS2-specific analyses. Both Kaggle sources likely over-represent notable repositories, introducing selection bias. Finally, repositories may not be entirely independent, projects within the same organization or language ecosystem may share structural characteristics, potentially violating the independence assumption underlying all models.

4.3 Future Directions

Several natural extensions follow from this work:

- **Mixed effects models (lme4):** Account for the nested structure of repositories within organizations or language communities.
- **Generalized Additive Models (GAMs):** Relax the log-linearity assumption and capture potential nonlinear relationships between forks and stars.
- **Longitudinal / time series models:** Track star growth trajectories over a repository's lifetime, incorporating creation date, commit history, and trending events.
- **Zero-inflated Negative Binomial:** Better model the long tail of near-zero-star repositories that are excluded from the current analysis.

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